

# Fractals

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# Outline

## Fractals Around Us

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Fractals Around Us

Fractals in Geometry

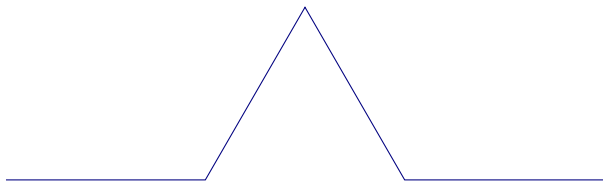
# Russian Dolls, Ferns, Trees, Cauliflower



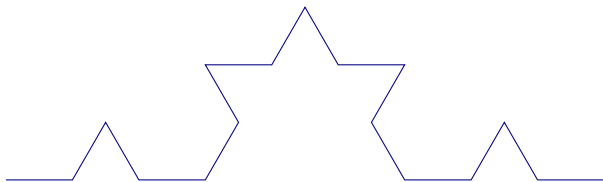
# Russian Dolls, Ferns, Trees, Cauliflower



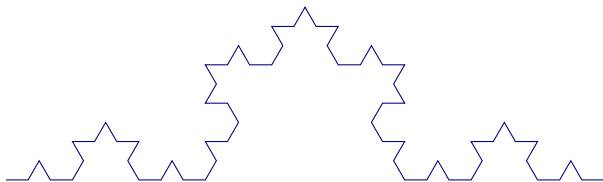
# Koch curve



# Koch curve

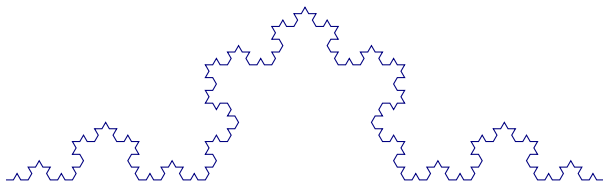


# Koch curve

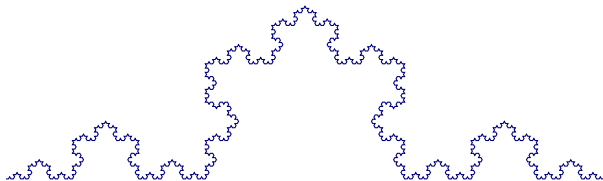




# Koch curve



# Koch curve



# Cantor Set



# Cantor Set



# Cantor Set



# Cantor Set

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# Cantor Set

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# Cantor Set

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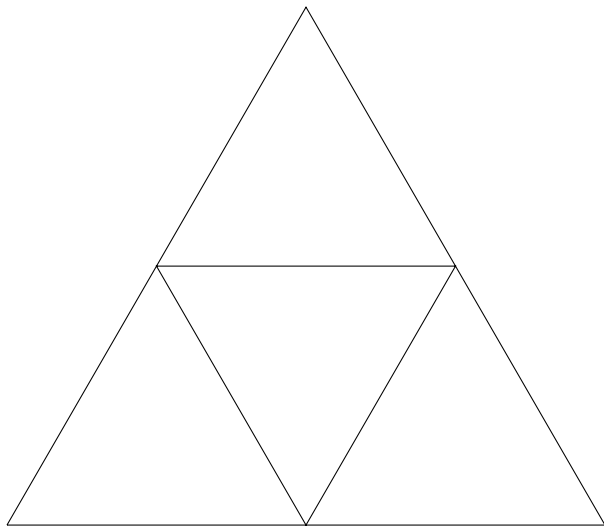
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00 00 00 00

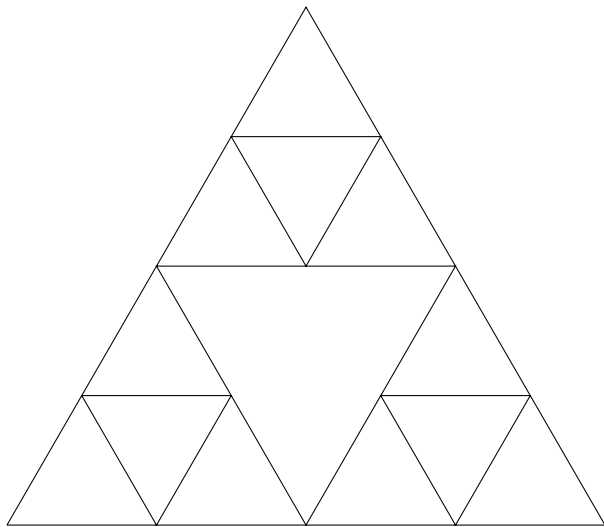
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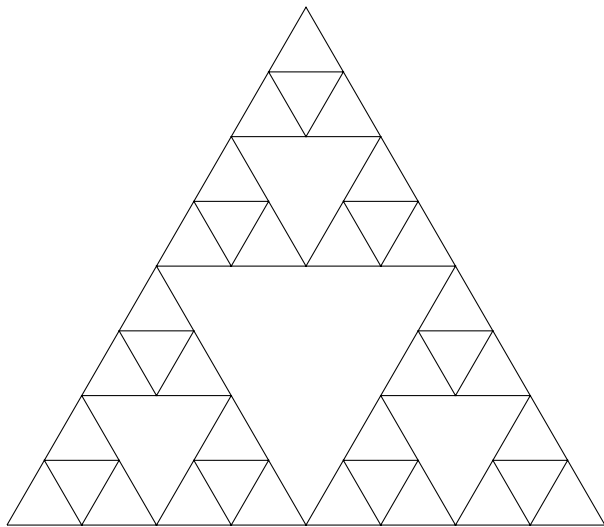
# Sierpinski Gasket



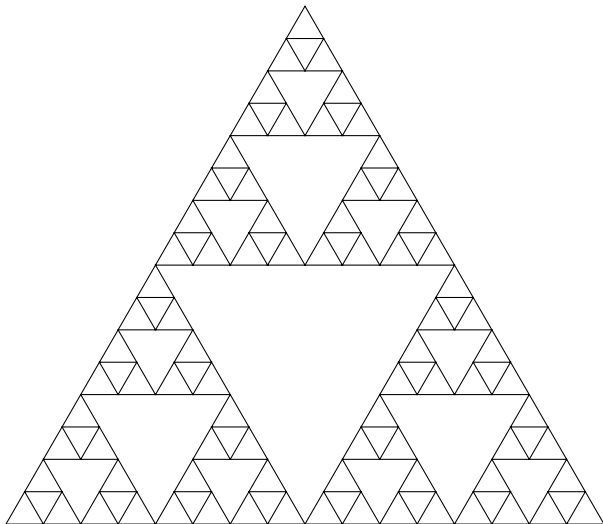
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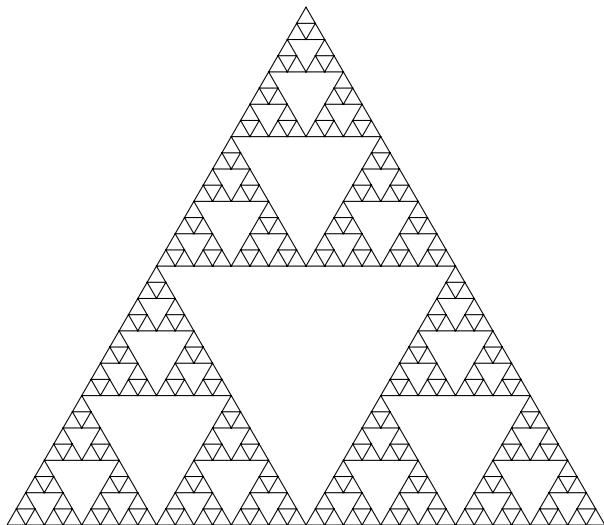
# Sierpinski Gasket



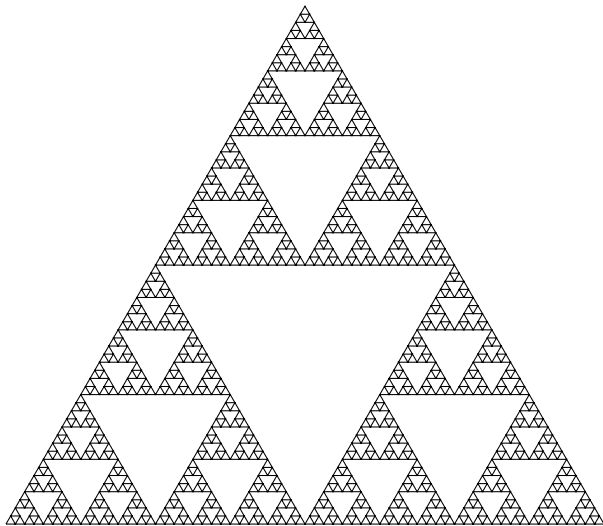
# Sierpinski Gasket



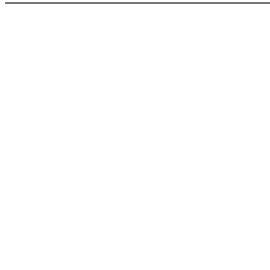
# Sierpinski Gasket



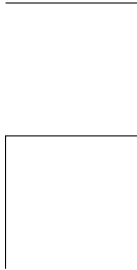
# Sierpinski Gasket



# Dragon Curve

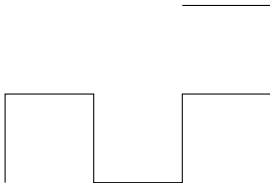


# Dragon Curve

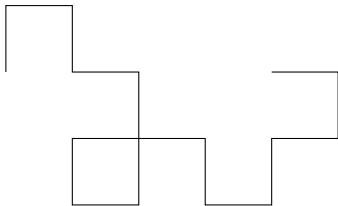




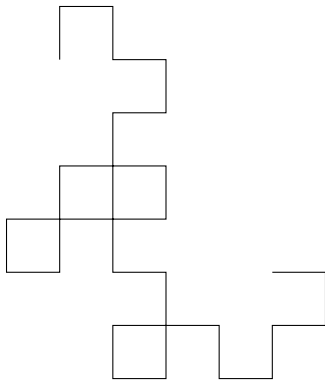
# Dragon Curve



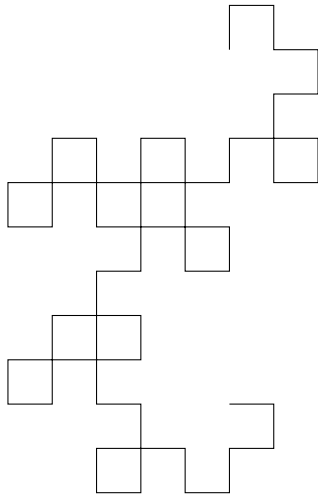
# Dragon Curve



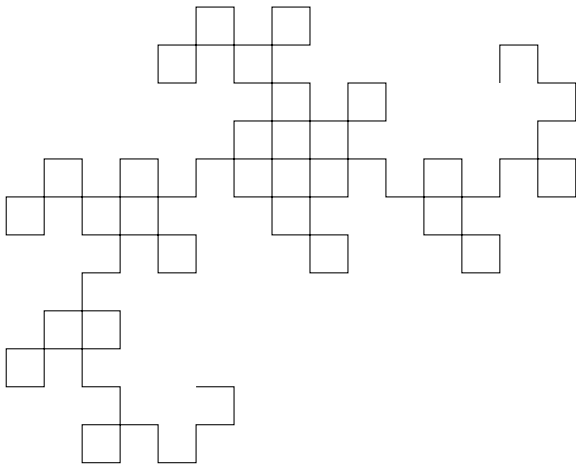
# Dragon Curve



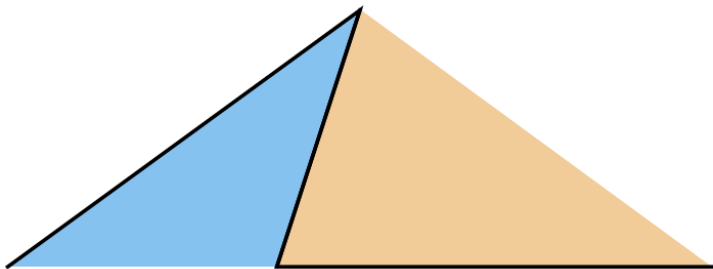
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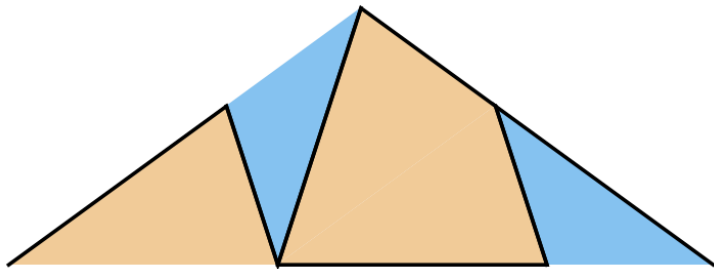
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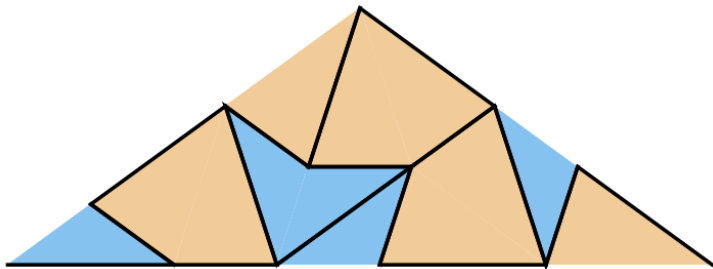
# Penrose Tiling



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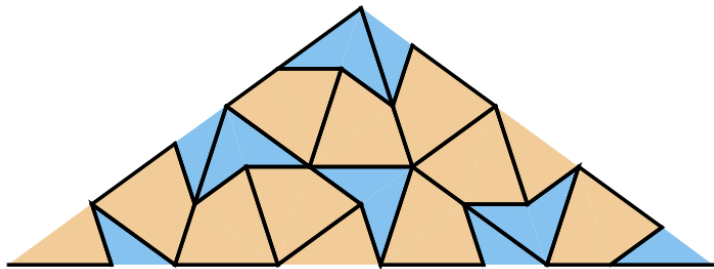


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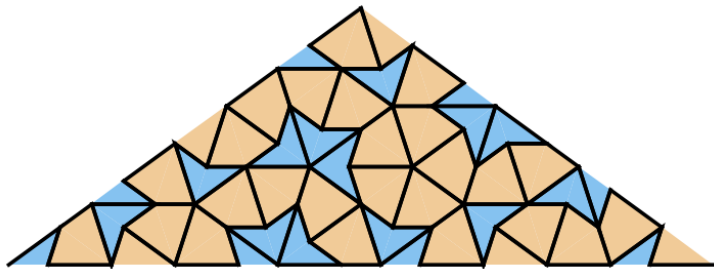




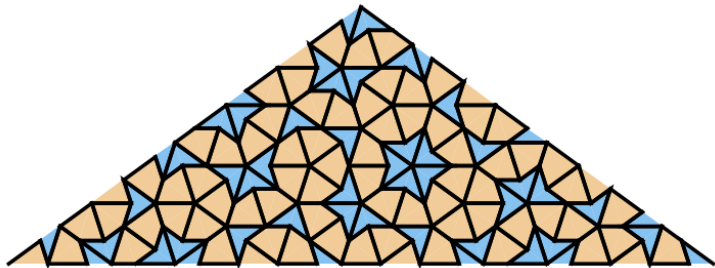
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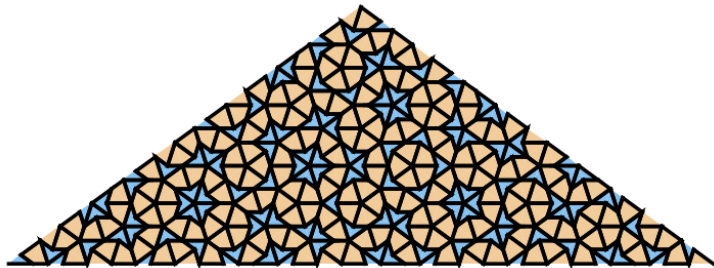
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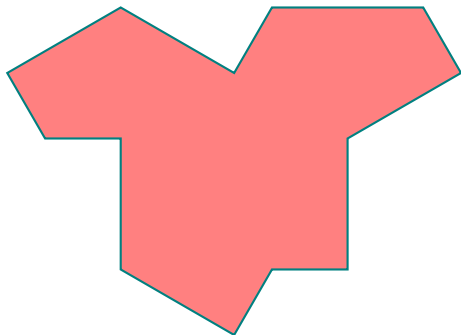


# Penrose Tiling



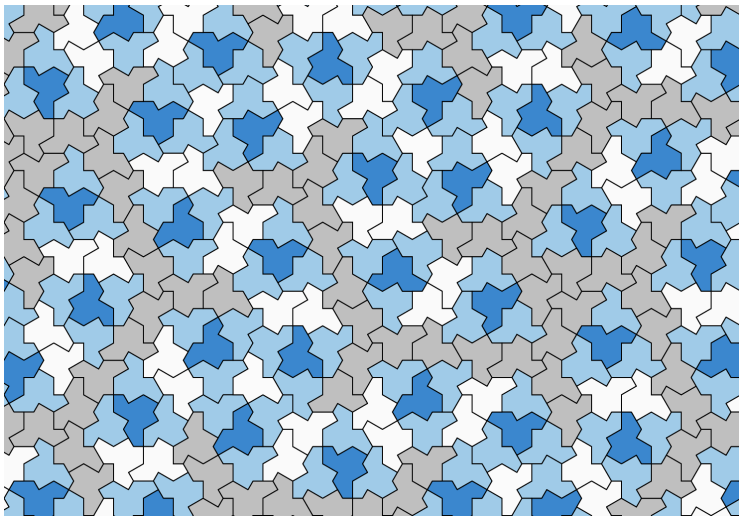
# Penrose Tiling



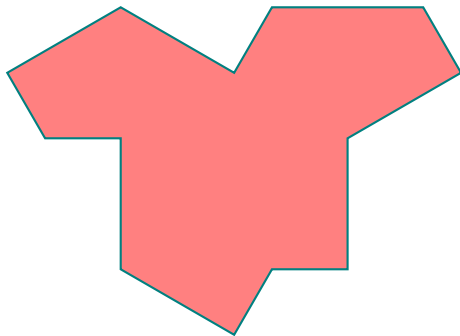


Hat polykite

# Einstein

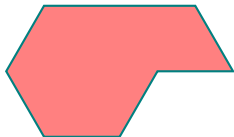


# Einstein



Tile( $1, \sqrt{3}$ )

# Einstein



Tile(1,0), Comet

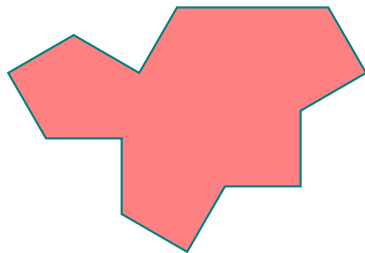


# Einstein



Tile(0,1), Chevron

# Einstein

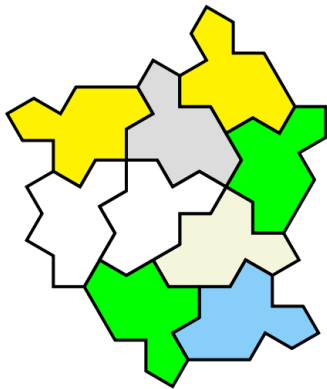


Tile(1,1)

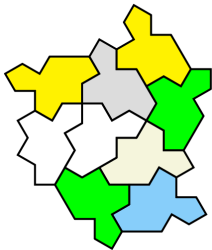
# Spectre



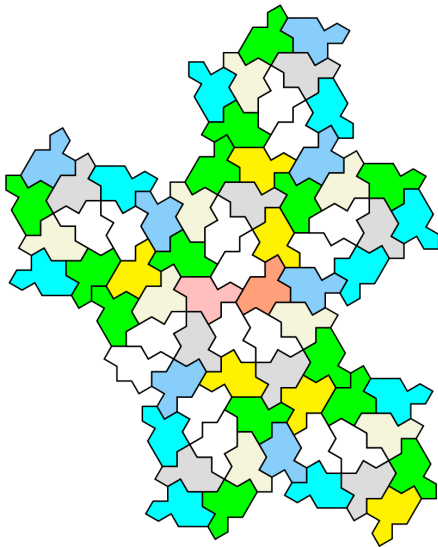
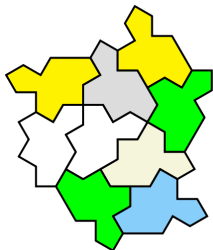
# Spectre



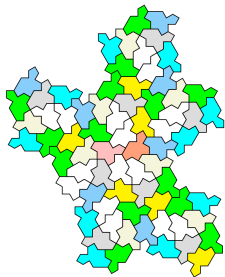
# Spectre



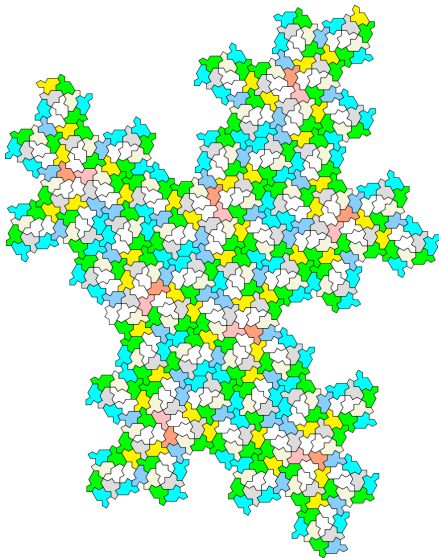
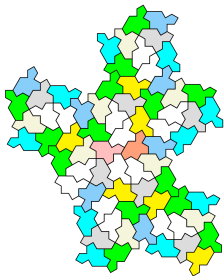
# Spectre



# Spectre

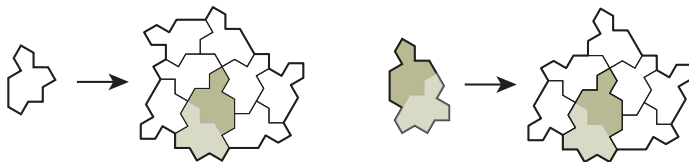


# Spectre





# Substitution Rules for Spectre



†

Replaces the Spectre by a cluster containing a Mystic and seven Spectres.

Replaces a Mystic by a cluster containing a Mystic and six Spectres

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†Smith, D., Myers, J.S., Kaplan, C.S. and Goodman-Strauss, C., 2023. *A chiral aperiodic monotile*. arXiv preprint arXiv:2305.17743

# References

Book 1. *The Fractal Geometry of Nature*, Benoît Mandelbrot

Article 2. *Penrose Tilings Tied up in Ribbons*, David Austin

Article 3. Smith, D., Myers, J.S., Kaplan, C.S. and Goodman-Strauss, C., 2023. *An aperiodic monotile*. arXiv preprint arXiv:2303.10798.

Article 4. Smith, D., Myers, J.S., Kaplan, C.S. and Goodman-Strauss, C., 2023. *A chiral aperiodic monotile*. arXiv preprint arXiv:2305.17743